# A Comprehensive Review of Artificial Intelligence for Weather Forecasting

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## ABSTRACT

The weather forecast is significant for people. It helps to make smarter daily decisions and stay out of harm's way. Accurate weather forecasting is one of the most difficult problems in the world. Artificial intelligence (AI) is becoming increasingly ubiquitous with applications in several fields finding solutions to some pressing and complex global problems. In recent years, many articles have been published regarding the applications of Machine learning and Deep learning to obtain forecasts. This article mainly discusses the different weather forecast models offered by different researchers. The focus is on work from 2009 to 2024. Finally, the main conclusion is that deep learning methods will be key elements of future weather forecasts.

## KEYWORDS

AI, Deep Learning, Machine Learning, Weather Forecasts.

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## 1 INTRODUCTION

Weather plays a huge role in human life. Many of our daily activities and actions depend on weather conditions. Additionally, due to unforeseen weather conditions, there is a huge loss of life and property. If we can effectively predict future weather conditions, we can prevent or reduce these losses. Artificial intelligence is emerging as one of the solutions to reduce weather-related losses, enabling weather analysis to provide weather forecasts that help improve benefits to society, including the protection of life and property, public health, and safety. In the literature, many machine learning and deep learning-based approaches have been explored in forecasting applications. Specifically, in the prediction of meteorological parameters. Research has been carried out to reduce the complexity of algorithms and improve the efficiency of the predictive power of models. The main question is What technique is most frequently used? Which algorithm has the best performance?

The main objective of this study is to review the Machine learning and Deep Learning techniques proposed in the literature for forecasting weather conditions. The technical steps taken by various researchers in this field have been reviewed and presented in this investigative document. In particular, articles published in international journals from 2009 to 2024 were considered. For the literature review stage, the following search engines for searching articles were widely used: ScienceDirect, Research Gate, and Google Scholar. The remainder of this article is structured as follows: Section 2 categorizes various approaches and algorithms used for forecasting meteorological parameters by various researchers in the present day. Section 3 brings out an analysis of the literature Survey. Section 4 presents a discussion of the results. Section 5 provides concluding remarks to summarize the paper.

## 2 RELATED WORKS

In the first study [1], the authors used RFR, SVR, MLPR, and ETR to predict the next day's temperature at any hour in the city of Nashville, Tennessee, based on data collected from wunderground.com. Evaluation results show that machine learning models can predict weather features with sufficient accuracy.

In the second study [2], the authors used SVR, and ANN to predict precipitation and temperature. This study concludes that SVR outperformed ANN in predicting precipitation, and ANN outperformed SVR in predicting temperature. The meteorological data used in this work was collected from the Meteorological Department of Bangladesh.

In another study [3], the authors evaluated the performance of three artificial intelligence techniques (KNN, ANN, and ELM) for rainfall forecasting in the state of Kerala in India. The ELM structure provides more accurate results for the JJAS and OND seasons than the KNN and ANN techniques in the currently used precipitation dataset. The time series data were collected from the Indian Institute of Tropical Meteorology.

In the literature [4], the authors compared several machine learning methods to predict the average daily and monthly rainfall of the city of Fukuoka in Japan. Their work suggests using a hybrid multi-model method coupled with classification and a selection of models to improve two of them. Rainfall data were collected from six nearby weather stations in Fukuoka and Saga prefectures in Japan from 1975 to 2009.


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The work of [5], presented an application of SVM for one-day maximum temperature prediction based on data from the previous n days. They also compared the output result with MLP. The results show that SVM performs better than MLP. Weather data from the University of Cambridge for five years is used to build this model.

In Ref. [6], the authors obtained the best 14 algorithms to build models to predict rainfall in Sudan. They concluded that the KStar algorithm has the maximum correlation coefficient, the minimum RMSE, and the third lower MAE. The meteorological data used in this work was imported from the Central Bureau of Statistics of Sudan 13 years ago.

In the following case study [7], the authors compared SDAE with standard artificial neural networks to predict air temperature from historical pressure, humidity, and temperature data collected from meteorological sensors in northwest Nevada. The results indicate that SDAE performs better than NN in the temperature prediction domain.

In another study [8], the authors developed an ANN model that can be used to predict daily mean ambient temperatures for Denizli, Turkey. The model was trained and tested using temperature data obtained from the Turkish State Meteorological Service over three years. The results show that the ANN technique can be used reliably in temperature estimations.

In the literature [9], the authors developed and implemented two models ANN and MLR to predict precipitation on an annual and monthly basis. The final results suggest that the ANN model has a better performance than the MLR model. The dataset that was used in this study includes daily precipitation and temperature measurements for Alexandria, Egypt.

In the work of [10], built an artificial neural network (FFNN) to predict rainfall. The result showed that the FFNN model can be used as a predictive algorithm for precipitation forecasting. Rainfall data were collected from the website www.iniawaterportal.org.

The authors of [11] the authors applied one of the ANN models, namely BPNN, to predict rainfall. The results of this study showed that BPNN models can be used as a predictive algorithm that provides good predictive accuracy. The rainfall data used was taken from Tenggarong station, East Kalimantan, Indonesia.

In Ref. [12], the authors predicted the daily maximum temperature based on SVMr. A comparison with alternative neural methods, based on statistical tests, showed that SVM outperforms a multilayer perceptron and an Extreme Learning Machine in this prediction problem. Actual data measured at European weather stations for 10 years was used.

In the following case study [13], the authors developed and applied ANN to predict maximum daily precipitation for the next coming year. The analysis results showed a quite satisfactory relationship between the respective maximum daily rainfall predicted and observed one year in advance. The data used for the analysis was recorded at the National Observatory in Athens.

In another study [14], the authors proposed an improved KNN algorithm for predicting precipitation. The result showed that the accuracy of our proposed precipitation forecasting approach based on the improved KNN algorithm is better than those of DWKNN, WKNN, and KNN in almost all test cases. They used daily rainfall data from June to August, which is collected from twenty nationwide automatic weather stations in Beijing.

In the literature [15], the authors predicted the maximum temperature of Tehran in winter based on ANN.

In the work of [16], the authors applied ANN to forecasting, and estimate monthly rainfall data belonging to the Zab watershed.

The authors of [17] the authors propose a method for temperature prediction using three machine learning models MLR, ANN, and SVM, the dataset used represents the weather conditions and variations of Central Kerala during the period 2007 to 2015. The error metrics and the CC show that MLR is a more precise model for temperature prediction than ANN and SVM.

In Ref. [18], the authors explore three machine learning models for weather prediction namely SVM, ANN, and RNN. the data was collected from different airport weather stations in India. This dataset contains data from the previous 12 years 2006-2018. It is found that Time Series-based RNN does the best job of predicting the weather.

In the following case study [19], used 5 different machine learning models which are ANN, SVR, DT, RFA, and LSTM to forecast rainfall. Average weekly rainfall data of the Kuala Krai rainfall station have been used as a predictor variable for this study. The results have shown that LSTM performed best among other models in forecasting the rainfall of Kuala Krai station.

Another study [20], applied the ANN for the prediction of heavy precipitation every month. For this purpose, Historical rainfall data was collected from 116 rain gauge stations for 50 years from 1965 to 2015. The results indicate that the ANN model is reliable in anticipating the risky level of heavy precipitation events.

In the literature [21], the ANN model was employed to forecast rainfall for Bangkok. Historical rainfall data was collected from 104 stations of the BMA and TMD rain gauge networks for the period from 1991 to 2005. The ANN model was found to be efficient in fast computation and capable of handling the noisy and unstable data that are typical in the case of weather data.

In the work of [22], the authors have utilized ANN modeling for the prediction of monthly rainfall in Mashhad synoptic station which is located in Iran. To achieve this model, monthly rainfall data from 1953 to 2003 for this synoptic station was used. The achieved ANN model gave satisfactory prediction performance.

In the literature [23], the authors develop a model using AI based on the ANN for the prediction of daily and monthly rainfall. The dataset was obtained from WeatherUnderground.com, at the Austin KATT station. The results show that the ANN model is a promising algorithm to predict daily/monthly rainfall.

Finally, the authors of [24] used ANN to predict the weather in the village of Delanggu, Klaten. it was found that the ANN model which had the smallest error and greatest accuracy used 40 training cycles and 7 hidden layers. The data was collected from the proposed IoT-based weather station located in Delanggu village in September 2021.

## 3 ANALYSIS

Table 1 explains the comparison of different methods and models in the literature by various authors on weather forecasting. The table is divided into Seven columns: Reference, Year, Input Data, Functionality, Region, Models, and Accuracy measure.

**Table 1: Analysis of different methods for weather forecasting.**

| Ref | Year | Input Data | Functionality | Region | Models | Error/Accuracy |
|---|---|---|---|---|---|---|
| [1] | 2018 | Temperature, humidity, wind direction, atmospheric pressure | next day's temperature | UNITED STATES | RFR, SVR, MLPR, ETR | RFR and ETR exhibit an almost similar RMSE close to 3.0 which is the lowest among the two previous models |
| [2] | 2017 | Rainfall, temperature | Precipitation and temperature forecast | Bangladesh | SVR, ANN | SVR has surpassed ANN in predicting rainfall with RMSE=27.6 and MAE=0.17. ANN surpassed SVR in temperature prediction with RMSE=8.4 and MAE=1.72 |
| [3] | 2018 | Precipitation | Predict rainfall during monsoon seasons (JJAS and OND) | India | KNN, ANN, ELM | The ELM technique presents the best results with MAE=3.075, RMSE=3.809 for the JJAS rainy season and with MAE=3.149, RMSE=6.000 for the rainy season OND |
| [4] | 2012 | daily and monthly precipitation | Forecast Average Daily and Monthly Precipitation | Japan | ANN, KNN, MARS, SVR, MM | For the daily rainfall, MM produced the most accurate forecast (RMSE=7.555). For the monthly rainfall, SVR produced the most accurate forecast (RMSE=28,815) |
| [5] | 2009 | Max temperature | Predict max temperature | England | SVM | MSE=7,07 |
| [6] | 2014 | Wind D, Humidity, Min-T, Max-T, Wind S | Predict precipitation | Sudan | GPs, LR, MLP, IBk, Kstar Additive Regression, Bagging, RS, Regression by Discretization, DT, M5Rules M5P, REPTree, UserClassifier | The Kstar technique presents the best results (RMSE=0.2285, MAE=0.1091) |
| [7] | 2015 | Temperature, pressure, humidity, wind speed | temperature forecast | UNITED STATES | SDAE | RMSE=2.06 |
| [8] | 2009 | Average daily temperature values | Predict average ambient temperatures | Turkey | ANN | RMSE=1.9655. |
| [9] | 2011 | Max and min temperatures, precipitation, wind speed. | Precipitation forecast on an annual and monthly basis | Egypt | ANN, MLR | ANN presents the best results with RMSE= (Yearly=32.04, January=2.83, December=10.04) |
| [10] | 2018 | Temperature, cloud cover, vapor pressure, precipitation | Predict precipitation | India | FFNN | Accuracy= 93.55%. |
| [11] | 2015 | Precipitation | Predict precipitation | Indonesia | BPNN | MSE=0.00096341 |
| [12] | 2011 | Min, average and max temperature, precipitation, atmospheric pressure, snow depth, sunshine duration, relative humidity, cloud cover | Daily max temperature prediction | Europe | SVMr | RMSE=2.8318 |
| [13] | 2014 | Max daily precipitation | Predict the max daily rainfall | Greece | ANN | RMSE=16.4 |
| [14] | 2017 | Daily precipitation | Precipitation forecast | China | KNN | Accuracy=49.46%. |
| [15] | 2019 | Tmax, Rtempt, n, U2, RH mean | Max temperature prediction | Iran | ANN | RMSE=0.104, MAE=0.311 |
| [16] | 2015 | Min and max relative humidity, min temperature, pressure, precipitation | Precipitation forecast | Iran | ANN | ANN of Piranshahr, (MSE=3.88, RMSE=1.97, MAE=1.13), ANN of Sardasht, (MSE=8.43, RMSE=2.90, MAE=1.68) |
| [17] | 2019 | Humidity, pressure, wind speed, wind direction, temperature | Temperature prediction | India | MLR, ANN, SVM | MLR is a more precise model than ANN and SVM. RMSE= (1.8568, 2.0100, 1.8727) MAE= (1.3777, 1.5174,1.3667) |
| [18] | 2019 | Air temperature, Atmospheric pressure, Relative humidity, Average wind direction, Total cloud cover, Horizontal visibility, Dew point temperature. | Predict temperature | India | SVM, ANN, RNN | RNN is better than SVM and ANN. RMSE= (6.67, 3.1, 1.41) |

| | | | | | | |
|---|---|---|---|---|---|---|
| [19] | 2020 | Average weekly rainfall | Rainfall forecast | Malaysia | LSTM, RFA, SVR, ANN, DT | LSTM performed best among other models, RMSE= (8.34, 8.41, 9.08, 9.10, 11.15), MAE= (4.83, 5.31, 6.81, 5.83, 7.07) |
| [20] | 2017 | Monthly rainfall data and past monthly rainfall | Prediction of heavy precipitation | Malaysia | ANN | Best result: Model C, RMSE= 0.06 |
| [21] | 2009 | Rainfall intensity, Relative humidity, wet bulb temperature, Air pressure, Cloudiness at t, Average hourly rainfall intensity of all rain gauges, Rainfall intensity at surrounding station 1, Rainfall intensity at surrounding station 2,Rainfall intensity at surrounding station 3 | Rainfall forecasts | Thailand | ANN | RMSE=0.71 |
| [22] | 2016 | average rainfall, last year rainfall,last five bi-monthly rainfall | Predict rainfall | Iran | ANN | M531 (RMSE= 0.99, MAE= 6.45), M741 (RMSE= 0.96, MAE= 5.97) |
| [23] | 2020 | Temperature, dew point, humidity, pressure, visibility, and wind speed | Predict daily monthly rainfall | United States | ANN | Daily rainfall (RMSE=0.2487, MAE=0.0932) Monthly rainfall (RMSE=0.0731, MAE=0.0578) |
| [24] | 2022 | max and min (Temperature, Humidity, wind speed, light intensity), wind direction, Rainfall | Predict the weather | Indonesia | ANN | Accuracy(Temp=90.5%, Hum=87.2%, Rain=68.2%, Wind Speed=89.6%, Light Intensity=83.1%) |

Table 1 presents a series of studies using algorithms for different meteorological parameter forecasting problems. Most studies have used neural networks and have concluded that past historical parameters have a greater impact on accuracy than the number of hidden layers. Moreover, this work demonstrates the importance of a preprocessing step, since simple models with well-chosen input variables tend to perform better.

## 4 DISCUSSION

In this section, the analysis of the results of recent work on the prediction of weather is discussed, specifically focusing on temperature and rainfall prediction which have been summarized in the tables below to clarify and compare the models of machine learning and deep learning.

Figure 1,2,3 and 4 show the comparison between models using four evaluation processes RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), MSE (Mean Squared Error), and ACCURACY which indicates the learning efficiency of the model. Among the quantitative indicators employed for assessing the performance of the models, RMSE, MAE, and MSE are negative indicators, which means that the lower values of which indicate better prediction accuracy. Comparing the performance of different forecasting models is a tricky task. however, this document has highlighted the main concerns of the various models discussed. The contrasts drawn between the different models, although based on different data sets and criteria, should help give future researchers a starting point to decide which type of approach to use based on their specific needs.

Fig.1 shows the comparison between different algorithms, based on RMSE. it can be seen that ANN(Artificial Neural Networks) reported lower RMSE values and hence exhibited better prediction accuracy in rainfall forecasting, also It can be perceived that RNN (Recurrent Neural Networks) reported minimum error values in temperature forecasting compared to other models. In Fig.2 we can see that ANN (Artificial Neural Networks) has a slightly lower MAE among other techniques in rainfall and temperature prediction. It is observed from Fig.3 that BPNN (Back Propagation Neural Network) reported a lower MSE value, demonstrating its superior accuracy in forecasting rainfall. It is visualized from Fig.4 that FFNN (Feed Forward Neural Network) reported the best prediction accuracy in rainfall forecasting, also ANN achieved the highest accuracy in predicting temperature.

The detailed survey concludes that Artificial Neural Networks (ANN) are the most commonly used and reliable deep learning architectures for weather prediction. These architectures offer a solution to complex nonlinear problems that are challenging to tackle using traditional techniques. ANNs can be utilized for weather forecasting due to their capability to uncover relationships among variables within datasets. Deep learning is recognized as a highly promising approach for weather forecasting. The studies confirm that deep learning is reliable for accurate weather forecasting, especially temperature and rainfall forecasting. Accurate and reliable weather forecasting plays a vital role in enabling informed decision-making and enhancing safety across all aspects of human life.

**Table 2: Machine learning algorithms and their performance criteria for weather forecasting.**

| Algo | Input Data | Output | Error/Accuracy |
|---|---|---|---|
| RFR, ETR | Temperature, humidity, wind direction, pressure | Next day's temperature | RMSE=3.0 |
| SVR | Rainfall, temperature | Precipitation | MAE=0.17 |
| MM SVR | daily and monthly precipitation | Average Daily and Monthly Precipitation | RMSE=7.555, RMSE=28.815 |
| SVM | Max temperature | Max temperature | MSE=7,07 |
| Kstar | Wind D, Humidity, Min-T, Max-T, Wind S | Precipitation | MAE = 0.1091 |
| SVMr | Min, average and max temperature, precipitation, atmospheric pressure at sea level, snow depth, sunshine duration, relative humidity and cloud cover | Daily max temperature | RMSE = 2.8318 |
| KNN | Daily precipitation | Precipitation | Accuracy=49.46% |
| MLR | Humidity, pressure, wind speed, wind direction, temperature | Temperature | MAE=1.3777 |

**Table 3: Deep learning algorithms and their performance criteria for weather forecasting.**

| Algo | Input Data | Output | Error/Accuracy |
|---|---|---|---|
| ANN | Rainfall, temperature | Temperature | MAE=1.72 |
| ELM | Precipitation | Rainfall during important monsoon seasons (JJAS and OND) | MAE=3.075, MAE=3.149 |
| SDAE | Temperature, pressure, humidity and wind speed | Temperature | RMSE=2.06 |
| ANN | Average daily temperature | Average daily ambient temperatures | RMSE=1.9655 |
| ANN | Max and min temperatures, precipitation, wind speed. | Precipitation forecast on an annual and monthly basis | RMSE=32.04, RMSE=2.83, RMSE=10.04 |
| FFNN | Temperature, cloud cover, vapor pressure and precipitation | Precipitation | Accuracy=93.55% |
| BPNN | Precipitation | Precipitation | MSE=0.00096341 |
| ANN | Max daily precipitation | Max daily rainfall | RMSE=16.4 |
| ANN | Tmax, Rtempt, n, U2, RH mean | Max temperature | MAE=0.311 |
| ANN | Min and max relative humidity, min temperature, pressure, precipitation | Precipitation | MSE=3.88, MSE=8.43 |
| RNN | temperature, pressure, Relative humidity, Average wind direction, Total cloud cover, Horizontal visibility, Dew point temperature | Temperature | RMSE=1.41 |
| LSTM | Average weekly rainfall | Rainfall | MAE=4.83 |
| ANN | Monthly rainfall data, past monthly rainfall | precipitation on monthly | RMSE=0.06 |
| ANN | Rainfall intensity, Relative humidity, wet bulb temperature, Air pressure, Cloudiness, Average hourly rainfall intensity of all rain gauges, Rainfall intensity at surrounding station, Rainfall intensity at surrounding station 2, Rainfall intensity at surrounding station 3 | Rainfall | RMSE=0.71 |
| ANN | average rainfall for the estimated mont, last year rainfall ,last five bi-monthly rainfall | Monthly rainfall | MAE=6.45, MAE=5.97 |
| ANN | Temperature, dew point, humidity, pressure, visibility, and wind speed | Daily and monthly rainfall | MAE=0.0932, MAE=0.0578 |
| ANN | max and min (Temperature, Humidity, Rainfall, wind speed , max and min light intensity, wind direction, light intensity | Weather for the next three days | Accuracy (Temp=90.5%, Rain=68.2%) |

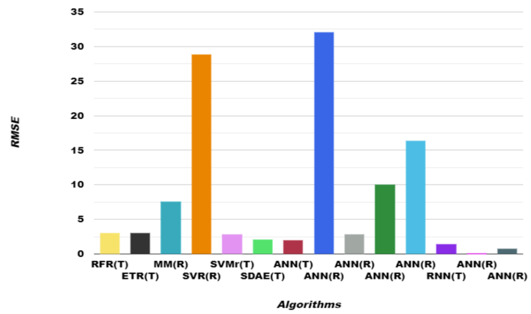


**Figure 1: Comparison between models using RMSE.**

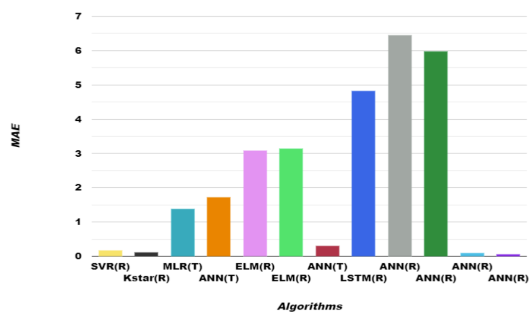


**Figure 2: Comparison between models using MAE.**

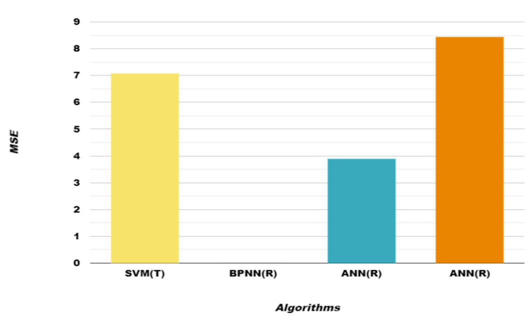


**Figure 3: Comparison between models using MSE.**

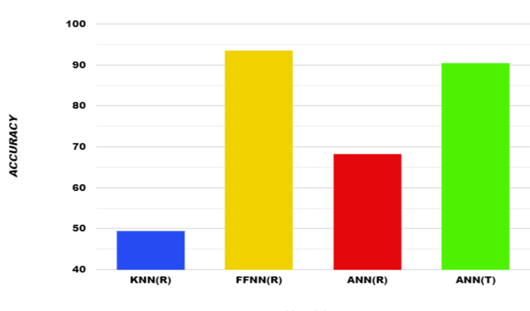


**Figure 4: Comparison between models using ACCURACY.**

## 5 CONCLUSION

Accurate forecasting of weather parameters is very important and fundamentally necessary, as early warnings of severe weather events can help prevent loss and damage caused by climate change. In this article, we proposed a survey that discusses the recent research works related to weather forecasting, offering a detailed analysis of the results.

The findings reported by the authors are analyzed to evaluate the performance of different models. ANN is established to be a more reliable deep learning technique for weather forecasting, offering promising results in the field of weather forecasting. This detailed literature review will serve as a reference tool for scholars interested in the topic of weather forecasting. Moreover, in the future, we plan to employ Artificial Neural Networks (ANNs) for the prediction of temperature and rainfall, with a focus on using the best-performing training algorithm and the optimal activation function.

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# ABBREVIATIONS

**AI** Artificial Intelligence
**RFR** Random Forest Regression
**SVR** Support Vector Regression
**ETR** EnergyTradeRank
**ANN** Artificial Neural Network
**KNN** K-Nearest Neighbour
**ELM** Extreme Learning Machines
**MARS** Multivariate Adaptive Regression
**MM** Multi-modèle hybride
**SVM** Support vector machine
**LR** Linear Regression
**MLP** Multi-Layer Perceptron
**IBK** Instance Based K-Nearest Neighbor
**RS** Random SubSpace
**DT** Decision Table
**SDAE** Stacked Denoising Autoencoder
**MLR** Multiple Linear Regression
**FFNN** Feed-forward neural network
**BPNN** Back Propagation Neural Network
**SVMr** Support Vector Machines Regression
**RNN** Recurrent Neural Network
**LSTM** Long Short-Term Memory
**RFA** Robust Federated Aggregation
**RMSE** Root Mean Square Error
**MAE** Mean Absolute Error
**MSE** Mean Squared Error